Homework 6

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Exercise 1

Let the game have the following payoff matrix (rows for the kicker, columns for the goalkeeper).

	Left	Center	Right
Left	$0.60, 0.40$	$0.95, 0.05$	$0.95, 0.05$
Center	$0.95, 0.05$	$0.00, 1.00$	$0.95, 0.05$
Right	$0.95, 0.05$	$0.95, 0.05$	$0.60, 0.40$

Note that the Left and Right strategies are identical; this indicates that the Nash equilibrium, the kicker has the mixed strategy $(p, 1-2p, p)$ and the kicker has $(q, 1-2q, q)$ for some $0 \leq p, q \leq 1$.

If the goalkeeper chooses left or right, the expected payoff for the kicker is:

$$0.60q + 0.95(1-2q) + 0.95q = 0.95 - 0.35q$$

and if the goalkeeper chooses center, it is:

$$0.95q + 0.00(1-2q) + 0.95q = 1.90q$$

In the Nash equilibrium, the payoffs are equal; this means that $q = \frac{19}{45} \approx 0.42$. The goalkeeper then has a mixed strategy $(0.42, 0.16, 0.42)$ in the Nash equilibrium.

If the kicker chooses left or right, the expected payoff for the goalkeeper is:

$$0.40p + 0.05(1-2p) + 0.05p = 0.05 + 0.35q$$

and if the kicker chooses center, it is:

$$0.05p + 1.00(1-2q) + 0.05q = 1.00 - 1.90q$$

In the Nash equilibrium, the payoffs are equal; this means that $p = \frac{19}{45} \approx 0.42$. The kicker then also has a mixed strategy $(0.42, 0.16, 0.42)$ in the Nash equilibrium.

If the kicker kicks into the left corner, the best response of the goalkeeper would to choose left as well.